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Ilex paraguariensis Extracts: A Source of Bioelements and Biologically Active Compounds for Food Supplements

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Abstract: *Ilex paraguariensis*, commonly known as yerba mate, is a plant belonging to the holly genus Ilex and the Aquifoliaceae family, indigenous to South America, and is used for the production of yerba mate. Yerba mate is renowned for its abundance of essential nutrients and bioactive compounds. Based on test results, it can be assumed that the selection of raw material for the preparation of extracts as well as the extraction method significantly influence the final content of biologically active compounds in the extracts. Consequently, this variability impacts the ultimate concentration of biologically active substances within the end product, potentially influencing human consumption. The present study aimed to quantify and compare the content of selected biological active compounds in supplements and products containing *I. paraguariensis* extracts, along with organic yerba mate dried through a smoke-free process, available in the European market (P-1–P-10). The evaluation focused on antioxidant substances such as neochlorogenic acid, chlorogenic acid, cryptochlorogenic acid, 4-feruloylquinic acid, isochlorogenic acid, rutoside astragalin, and caffeine. Additionally, the concentration of specific macro and trace elements was ascertained. The antioxidant compound makeup differs between methanol-extracted samples and aqueous extracts. In both cases, methanol extracts, particularly those in instant and traditional herb forms, showed the highest content of organic compounds with antioxidant properties (such as phenolic compounds and caffeine). The highest content of chlorogenic acid was detected in both methanol (14.7412 mg/g d.w.) and water (8.3120 mg/g d.w.) extracts in product P-4. The caffeic acid content ranged from 0.1491 mg/g d.w. to 1.7938 mg/g d.w. in methanol extracts and from 0.0760 mg/g d.w. to 0.4892 mg/g d.w. in water extracts. The neochlorogenic acid content ranged from 2.6869 to 23.9750 mg/g d.w. in ethanol extracts and from 0.4529 to 10.2299 mg/g d.w. in water extracts. Therefore, the traditional preparation of yerba mate as a water infusion does not fully exploit the raw material's potential. Among the tested products, only the dietary supplement in capsule form contained protocatechuic acid, which was not present in any other tested products. Conversely, compounds characteristic of yerba mate found in other preparations were absent in this supplement. The caffeine content was also the lowest in this product. The determined content of active substances did not consistently match the declarations made by producers if stated on the packaging.

Keywords: yerba mate; *Ilex paraguariensis*; dietary supplements; caffeine; bioelements; antioxidants; yerba mate extract; chlorogenic acids

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1. Introduction

In contemporary times, people's awareness of the impact of food and dietary supplements on human health is significantly increasing. The results of research on the content of

biologically active compounds in food can be used in the rational planning of the daily diet and in the production of functional food [1]. *Ilex paraguariensis*, thanks to its unique content of phenolic compounds, especially chlorogenic acid and its derivatives, has great potential to play an important role in the prevention of lifestyle diseases [2–4]. *I. paraguariensis* is an indigenous flora predominantly found in the regions of Argentina, Paraguay, Brazil, and Uruguay. This species, from the Aquifoliaceae family, is an evergreen tree or shrub. The leaves and twigs of this plant serve as the primary constituents for the production of yerba mate, a traditional beverage [2]. As a part of the daily diet for millions of people, mainly in South America [2], in the form of the yerba mate beverage, it is widely consumed. Extracts from *I. paraguariensis* are increasingly used as an ingredient in dietary supplements, including those with stimulating or slimming effects.

Historically, this plant has been utilized for medicinal purposes since the pre-Columbian era, with indigenous populations employing it as a means to alleviate sensations of hunger, fatigue, and stress [5,6]. The monograph for the leaves of this plant, "*I. paraguariensis* St. Hilaire, folium", has been included in the European Pharmacopoeia since 2018 [7]. Contemporary scientific investigations have attributed the biological properties of *I. paraguariensis* to the presence of biologically active compounds within the plant matter. These include amino acids, purine alkaloids, polyphenols, terpenoids, saponins, vitamins, and bioelements. From a health enhancement perspective, phenolic compounds constitute the most crucial category of substances for their antioxidant properties [8,9]. Quantitative analysis has determined the total phenolic content (TPC) to be approximately 51 mg/g of the plant's dry weight [10]. Among the 46 polyphenolic compounds identified in the tested extracts, the most abundant were chlorogenic acids: 3-caffeoylquinic acid (26.8–28.8%), 5-caffeoylquinic acid (21.1–22.4%), 4-caffeoylquinic acid (12.6–14.2%), and 3,5-dicaffeoylquinic acid (9.5–11.3%) [11,12].

The invigorating attributes of the beverage are attributed to the presence of methylxanthines, a class of purine alkaloids encompassing caffeine, theobromine, and theophylline. The dried extract of *I. paraguariensis* leaves contains a caffeine content ranging from 1% to 2% of the dry matter. The caffeine concentration in a single serving of holly infusion (approximately 150 mL) is estimated to be around 78 mg, a value that closely parallels the caffeine content in coffee, which contains approximately 85 mg in an equivalent volume. It is noteworthy that in the context of traditional yerba mate brewing practices, the volume of the infusion can reach up to 500 mL, and this quantity contains about 260 mg of caffeine or potentially more [8,10,13]. Yerba mate is also recognized as a source of rutin and astragalin. The isolation of these compounds in substantial quantities, in comparison to other nutraceuticals containing caffeine, suggests the existence of additional mechanisms of action and a beneficial impact on human health [13,14].

Scientific investigations have demonstrated that the bioactive compounds identified in extracts of *I. paraguariensis* leaves predominantly possess anti-inflammatory, antioxidant [8], antimicrobial [15], and anticancer properties [16–18]. These properties suggest potential applicability in the management of metabolic disorders, encompassing the regulation of body weight and diabetes [19–24]. Yerba mate consumption has a protective effect on cardiovascular health and significantly improves blood lipid profiles [25–27]. Empirical evidence indicates that the habitual consumption of infusions derived from this plant enhances cognitive functionality, augments cerebral blood flow, mitigates the risk of depression, reduces Aβ oligomer activities, which are thought to be primary contributors to the onset of Alzheimer's disease, shows neuroprotective effects and may play a significant role in the prophylaxis of Parkinson's disease [28,29].

The traditional yerba mate production process involves drying the raw material with smoke, which may affect the presence of polycyclic aromatic hydrocarbons [30,31]. Currently, the market offers a variety of yerba mate products that are hot air-dried, smoke-free, and sourced from organic farming as well as instant forms of yerba mate, powders, and supplements. It seems reasonable to consider the potential of *I. paraguariensis* as a

source of bioactive compounds for the development of new products intended for the pharmaceutical, nutraceutical, and food sectors [32,33].

Research conducted by Rząsa et al. (2022) has demonstrated that the geographical origin and production conditions of yerba mate significantly influence not only the taste, aroma, and quality of the beverage but also the quantity of biologically active substances in the final product. Thermal treatment, such as roasting, can deplete yerba mate of organic compounds possessing antioxidant properties [13]. It has been shown that the production process significantly affects the content of phenolic compounds in the finished yerba mate product, and thus its antioxidant activity. Therefore, to preserve these beneficial antioxidants, it is crucial to employ suitable preparation methods for these products. Based on the outcomes of the aforementioned research, it can be postulated that the choice of raw material for extract preparation will substantially affect the final content of biologically active compounds. Extraction conditions also play a key role in maintaining their stability; research indicates that microwave-assisted extraction seems to be the most effective [34–37]. Traditional yerba mate beverages are prepared as water extracts. For chimarrão, 85 g of yerba mate (smooth, traditional, native, coarse ground) is brewed with 150 mL of water at 75 °C. For tereré, 50 g of yerba mate is extracted with 180 mL of cold water (11 °C) [2]. The typical method of consuming this drink involves repeatedly pouring extra water into the yerba mate. Various extraction methods used for identifying and quantitating active compounds in yerba mate include hot water extraction, cold water extraction, ethanol extraction, liquid carbon dioxide extraction, supercritical fluid extraction, ultrasonic extraction or hydrophilic natural deep eutectic solvent [2,38–40].

The current study represents a continuation of research efforts aimed at quantitatively determining selected biologically active compounds, including polyphenols, rutoside, and caffeine, as well as crucial elements, in dietary supplements and other products containing *I. paraguariensis* extracts, as well as in selected smoke-free yerba mate from organic farming. Subsequently, an evaluation was conducted to determine the contribution to meeting the daily demand for elements and phenolic compounds by analyzing products as well as the amount of caffeine intake.

2. Materials and Methods

2.1. Materials

The dietary supplements and products containing *I. paraguariensis* A.St.-Hil. from the Aquifoliaceae family extracts, as well as yerba mate obtained through a special manufacturing process (drying without smoke), used in the study have a commercial origin. The experimental yerba mate samples were stored in the Department of Inorganic and Analytical Chemistry at the Medical College of Jagiellonian University. In order to maintain confidentiality, the products being examined were designated as P-1 to P-10 (Table 1).

Table 1. Description of selected products.

Material	Form	Extract/Dosage		Country of Origin/Production	Cultivation	Organs of Plant
P-1 Food Supplement	Capsules	550 mg/45.1 mg polyphenols	1 × 1	Poland	Conventional	Leaf extract
P-2 Food Supplement	Capsules	480 mg/39 mg caffeine	1 × 1	England	Conventional	Leaf extract
P-3 Food Supplement	Extract powder	500 mg/40 mg caffeine	1 × 500 mg	Czech Republic	Conventional	Leaf extract
P-4 Yerba mate soluble	Instant	-	2 g	Paraguay	Conventional	-

Table 1. *Cont.*

Material	Form	Extract/Dosage		Country of Origin/Production	Cultivation	Organs of Plant
P-5 Yerba mate soluble	Instant	-	1 tablespoon	Brazil	Conventional	-
P-6 Yerba mate powder	Powder	-	1 g	Brazil/Poland	Conventional	-
P-7 Yerba mate	Tea bag	-	Tea bag—3 g	Brazil/Poland	Organic	95% large leaves and max 5% small sticks
P-8 Yerba mate despalada	Dry herbs	-	30–50 g (10–15 g for beginner)	Brazil/Poland	Conventional	95% leaves and max 5% small sticks
P-9 Yerba mate with extras Food Supplement	Dry herbs	-	30–50 g (10–15 g for beginner)	Brazil/Poland	Conventional	Leaves
P-10 Traditional yerba mate	Dry herbs	-	30–50 g (10–15 g for beginner)	Argentina	Conventional	Leaves and sticks

2.2. Reagents

Suprapur® nitric acid—65% and Suprapur® hydrogen peroxide—30% were purchased from Merck (Darmstadt, Germany). Fourfold distilled water with a conductivity below 1 μS/cm was obtained using the S2–97A2 distillation apparatus (Chemland, Stargard Szczecinski, Poland). Standards of Zn(II), Fe(III), Mn(II), Cu(II), and Mg(II) at a concentration of 1 g/L were purchased from the District Office of Measures (Łodz, Poland). The chemicals employed in this study were as follows: methanol (MeOH) and glacial acetic acid of analytical grade were procured from Chempur (located in Gliwice, Poland); high-performance liquid chromatography (HPLC)-grade MeOH was obtained from Merck (based in Darmstadt, Germany). The following standards were purchased: caffeic acid, chlorogenic acid, neochlorogenic acid, and rutoside from Sigma-Aldrich (St. Louis, MI, USA); isochlorogenic acid, 4-feruloylquinic acid and astragalin from ChromaDex (Los Angeles, CA, USA); and caffeine from Fluka Chemie AG (Buchs, Switzerland).

2.3. Determination of Tested Elements by Flame Atomic Absorption Spectroscopy (F-AAS)

The samples underwent mineralization before F-AAS analysis. For food supplements P-1 and P-2, the powdered extract was poured from the capsules, weighed, and homogenized. In other cases, after homogenization, the desired portions of the product were weighed. Three approximately 0.4 g samples of each product were weighed with 0.1 mg accuracy. The samples were then transferred to Teflon vessels, and 1 mL of 30% H_2O_2 solution and 5 mL of 65% HNO_3 were added. The wet mineralization process was carried out in a Magnum II microwave apparatus (ERTEC, Wrocław, Poland) in two stages of 10 and 20 min, at a power of 80% and 100%, with a pressure range of 42–45 Ba, maintaining the temperature at 295–300 °C. After mineralization, the solutions were transferred to quartz evaporators and heated for approximately 60 min at 140 °C on a hot plate to remove excess reagents to the so-called "almost dry" stage. The residues after evaporation were quantitatively transferred to 25-mL Erlenmeyer flasks and diluted up to the mark with fourfold distilled water. Samples were refrigerated until F-AAS analysis using an iCE 3000 Series Atomic Absorption Spectrometer (Cambridge, UK) equipped with a flame atomizer and SOLAAR Software, Version 2.01 software (Thermo Fisher Scientific, Waltham, MA,

USA). Calibration curves were determined for each analyzed element, and then its content in the analyzed solutions was determined. The parameters of the applied method were previously described using the same methodology and apparatus [41,42]. The obtained concentrations were calculated based on the average weight of the samples, taking into account mean masses for capsules. The results are expressed in mg/kg.

2.4. Determination of Organic Compounds in Yerba Mate Products by High-Performance Liquid Chromatography (HPLC)

2.4.1. Water Extraction

The brewing procedure usually requires water temperatures ranging from approximately 85 to 95 °C. Therefore, extracts from investigated products were prepared as follows: Approximately 2 g of investigated material from each product was weighed, and 60 mL of water at a temperature between 80 and 85 °C was introduced to the specimen. The extract was subsequently filtered twice post brewing (20 min). The presence of bioactive ingredients was tested in all extracts.

2.4.2. Methanol Extraction

The yerba mate material, once freeze-dried, was ground to a uniform consistency using an agate mortar. Subsequently, 4 g of this homogenized material was transferred to a 150 mL glass beaker and combined with 125 mL of analytical-grade methanol. An ultrasonic bath (Polsonic, Warszawa, Poland) was used for a 20 min extraction process. The resulting extracts were then filtered through paper filters into 300 mL crystallizers and left at room temperature (23 ± 2 °C) to allow the methanol to evaporate. This procedure was repeated six times to ensure thorough extraction of the raw material. Once the methanol had evaporated from the crystallizers, the remaining dry residue was dissolved in HPLC-grade methanol and filtered using membrane filters (ChemLand, Stargard, Poland). These extracts were then stored in a refrigerator. For chromatographic analyses, 1.5 mL of each extract was filtered using 0.22 µm PTFE syringe filters (ChemLand, Stargard, Poland) and transferred into glass vials (Witko, Łódź, Poland). Each sample was analyzed three times for accuracy.

2.4.3. Analysis of Phenolic Acids Flavonoids, and Caffeine

The RP-HPLC-DAD method was employed to determine the levels of phenolic acids, flavonoids, and caffeine in the materials under examination. The liquid chromatography equipment used in this study consisted of an HPLC analyzer (Merck Hitachi, Tokyo, Japan), a DAD-L2455 detector, an L-2350 thermostat, an L-2130 pump, a 4 × 250 mm RP-18 column (LiChrosfer, particle size 5 µm), and an L-2200 autosampler. The content of the analyzed substances was analyzed using gradient elution. Two solvents (A and B) were used as the mobile phase. Solvent A consisted of methanol and a 0.5% acetic acid solution in a 1:4 volume ratio, whereas solvent B was methanol. The following gradient was established: 100:0 for 0–25 min, 70:30 for 35 min, 50:50 for 45 min, 0:100 for 50–55 min, and 100:0 for 57–67 min. The flow rate was 1 mL/min, and measurements were carried out at 254 nm.

3. Statistical Analysis

Each sample was prepared and examined three times. The outcomes were presented as the mean ± standard deviation (SD) in percentage form. Data analysis was conducted using Microsoft Office Excel for Windows. Statistical analysis was performed using the Kruskal–Wallis test with Dunn's post hoc test to analyze statistically significant variances in the concentration of organic compounds in yerba mate. After categorizing the bioelements into product groups, a one-way analysis of variance was conducted for statistical analysis. A post hoc Tukey's HSD test was used to identify statistically significant differences between the product groups. A p-value of less than 0.05 was considered to indicate significant differences between groups. Data analysis was performed Statsoft STATISTICA v.14 software (Tulsa, OK, USA).

4. Results and Discussion

4.1. Metals Determined in Analyzed Products by Atomic Absorption Spectrometry

The quantification of metal content in the examined products containing *I. paraguariensis* extracts was accomplished through the application of flame atomic absorption spectrophotometry (F-AAS). This analytical technique is extensively employed for metal analysis. Table 2 shows the mean concentration of the studied macronutrient (Mg) and micronutrients (Cu, Fe, Mn, and Zn) in analyzed products P-1–P-10.

Table 2. Concentrations of analyzed metals in P-1–P-10 products [mg/kg]. Each analysis was conducted three times. The letters next to values represent Tukey's HSD post hoc results, different letters (a–g) indicate significant differences between homogeneous groups of yerba mate products for each bioelement ($p < 0.05$).

Product	Fe	Zn	Cu	Mn	Mg
P-1	66.1 ± 2.5 ^{bc}	5.7± 0.6 ^{f}	-	58.7± 3.3 ^{e}	4147.8 ± 558.4 ^{fg}
P-2	9.8 ± 0.3 ^{d}	22.3 ± 0.2 ^{e}	-	195.2 ± 8.2 ^{e}	2019.9 ± 116.1 ^{h}
P-3	-	20.7 ± 0.6 ^{e}	0.8 ± 0.1 ^{d}	114.3 ± 6.9 ^{e}	3387.9 ± 148.3 ^{g}
P-4	26.2 ± 3.8 ^{d}	139.4 ± 7.1 ^{a}	2.5 ± 0.2 ^{d}	993.4 ± 9.2 ^{c}	7746.8 ± 143. 8 ^{b}
P-5	14.8 ± 0.7 ^{d}	43.1 ± 0.7 ^{d}	-	1086.2 ± 21.9 ^{c}	9946.8 ± 383.4 ^{a}
P-6	207.5 ± 23.1 ^{a}	42.0 ± 1.5 ^{d}	5.2 ± 0.1 ^{c}	1498.9 ± 81.6 ^{b}	6315.0 ± 241.1 ^{cd}
P-7	107.1 ± 10.1 ^{b}	40.5 ± 2.7 ^{d}	9.9 ± 0.7 ^{a}	2897.8 ± 103.9 ^{a}	4978.1 ± 229.5 ^{ef}
P-8	91.1 ± 4.4 ^{c}	48.9 ± 2.1 ^{cd}	7.9 ± 0.9 ^{b}	1131.4 ± 75.6 ^{c}	6773.3 ± 507.0 ^{c}
P-9	90.9 ± 4.3 ^{bc}	53.6 ± 3.8 ^{c}	7.9 ± 0.7 ^{b}	1511.7 ± 25.8 ^{b}	5796.0 ± 176.5 ^{de}
P-10	180.2 ± 13.1 ^{a}	93.0 ± 5.6 ^{b}	7.7 ± 0.2 ^{b}	843.5 ± 29.5 ^{d}	5465.2 ± 145.6 ^{de}

The concentration of Mg determined in the tested products ranged from 2019 mg/kg for P-2 to 9946 mg/kg for P-5. The content of micronutrients was determined as follows: Zn from 5.7 mg/kg (P-1) to 139.4 mg/kg (P-4); Fe from 0 (P-3) to 207.5 mg/kg (P-5); Mn from 58.7 mg/kg (P-1) to 2898 mg/kg (P-7) and Cu from 0 (P-1, P-2, P-5) to 7.9 mg/kg (P-9). The content of elements varies statistically depending on the product. The highest contents of the tested elements were expected in extracts included in dietary supplements. In practice, however, product P-4, which is yerba mate soluble (instant), turned out to be the best source of Mg and Zn.

Statistically significant differences have been determined for the following groups of products: supplements P-1, P-2, P-3; instant products P-4, P-5; raw yerba mate P-7, P-8, P-9, P-10; and yerba mate powder P-6. There were no statistically significant differences in the rates of Mn concentrations between raw yerba mate and yerba mate powder ($p < 0.05$). In general, the following products had a significantly higher content of Fe, Zn, Cu, and Mg: instant products, raw yerba mate, and yerba mate powder, whereas dietary supplements containing extracts of *I. paraguariensis* (P-1, P-2, P-3) were characterized by a lower content of the tested elements.

Raw materials after the mineralization process had very high concentrations of the investigated elements, but in reality, there are differences in concentrations of these elements in raw materials and beverages after yerba mate preparation that need to be taken into account. Yerba mate contained a high concentration of Mg and Zn in comparison with other popular drinks such as coffee or tea, and significant amounts of them were found in beverages [13]. Based on the findings of bioelement content in the analyzed products and taking into account the dietary reference intakes (DRIs, mg/d) for these bioelements for males (31–50 years): Mg, 420; Cu, 0.9; Fe, 8; Mn, 2.3; and Zn, 11, and females (31–50 years): Mg, 320; Cu, 0.9; Fe, 18; Mn, 1.8; and Zn, 8, and the elements' concentrations determined, it can be concluded that only yerba mate brewed prepared according to the traditional

method using the usual 30–50 g of the product could be an important source of these microelements in the daily diet [43].

4.2. Analysis of Organic Compound Concentrations

Phenolic acids and other organic compounds were quantified using the RP-HPLC technique. The concentrations of organic compounds in the analyzed yerba mate products were determined after water and methanol extraction. An exemplary chromatogram of the analyzed sample of yerba mate products is shown in Figure 1. The obtained extracts of the tested yerba mate products (P-1–P-10) were analyzed. The chromatograms for samples P-2–P-10 revealed the presence of neochlorogenic acid, chlorogenic acid, 4-feruloylquinic acid, caffeic acid, isochlorogenic acid, caffeine, rutoside, and astragalin. However, the chromatograms for sample P-1, in both water and methanol extracts, did not show substances characteristic of yerba mate; only protocatechuic acid and caffeine were identified. The contents of the following antioxidant compounds were assessed: neochlorogenic acid, chlorogenic acid, 4-feruloylquinic acid, caffeic acid, and isochlorogenic acid. The findings were calculated per gram of dry weight and are presented in Tables 3 and 4. The structural formulas of the given compounds are shown in Figure 2. For the P-1 product, a chromatogram was obtained that lacked the phenolic constituents typically found in yerba mate but exhibited a distinct peak corresponding to an identified compound—protocatechuic acid.

In the analysis, the highest content of chlorogenic acid was detected in both methanol (14.7412 mg/g d.w) and water (8.3120 mg/g d.w) extracts in product P-4. The caffeic acid content ranged from 0.1491 mg/g d.w. and 0.0760 mg/g in methanol and water extracts to 1.7938 mg/g d.w. and 0.4892 mg/g d.w. in methanol and water extracts. The neochlorogenic acid content ranged from 2.6869 to 23.9750 mg/g d.w. in ethanol extracts and from 0.4529 to 10.2299 mg/g d.w. in water extracts. The composition of antioxidant compounds in methanol-extracted samples differs from that of water extracts. In general, the content of chlorogenic acids was significantly higher in methanol extracts compared to water extracts. Phenolic compounds in yerba mate exhibit antioxidant properties, contribute to overall health, and may reduce the risk of chronic diseases [44]. Phenolic compounds found in yerba mate, such as chlorogenic acids, could potentially serve as prebiotics. Chlorogenic acids may specifically influence the intestinal microflora, particularly the ratio of *Bacteroides* to *Firmicutes*. An imbalance in these two types of bacteria has been associated with the development of insulin resistance and obesity. Therefore, the consumption of yerba mate could potentially have beneficial effects on gut health and overall metabolic function [45,46].

Although there is no universally defined daily requirement for phenolic compounds, incorporating foods rich in these bioactive molecules contributes to health. Yerba mate, with its abundant phenolic content, can be a valuable addition to a balanced diet. For specific recommendations, the Academy of Nutrition and Dietetics suggests 400–600 mg of flavanols per day to support cardiometabolic health, but individual needs may vary, and a diverse diet ensures a broader intake of these beneficial compounds [47].

Additionally, the concentration of the following compounds was analyzed: caffeine, rutoside, and astragalin. The content of these compounds in P-1–P-10 products is presented in Table 5. The differences in the content of the tested compounds in ethanol and water extracts are additionally illustrated in Figure 3. Yerba mate is perceived as an energizing or relaxing drink, the consumption of which is similar to tea and infusions; yet there is limited awareness about its antioxidant properties [48]. Caffeine plays a pivotal role in conferring stimulant properties; therefore, the caffeine content was also determined. The content of this purine alkaloid ranges from 0.0278 to 1.1399 mg/g d.w. in water extracts and from 0.1963 to 3.4059 mg/g d.w. in methanol extracts. In a typical 150 mL serving of yerba mate infusion, the caffeine content is approximately 78 mg, which is quite similar to the caffeine found in coffee (around 85 mg). However, it is important to note that during traditional

yerba mate preparation, the beverage volume can occasionally extend to 500 mL, leading to an estimated caffeine content of 260 mg or more [8].

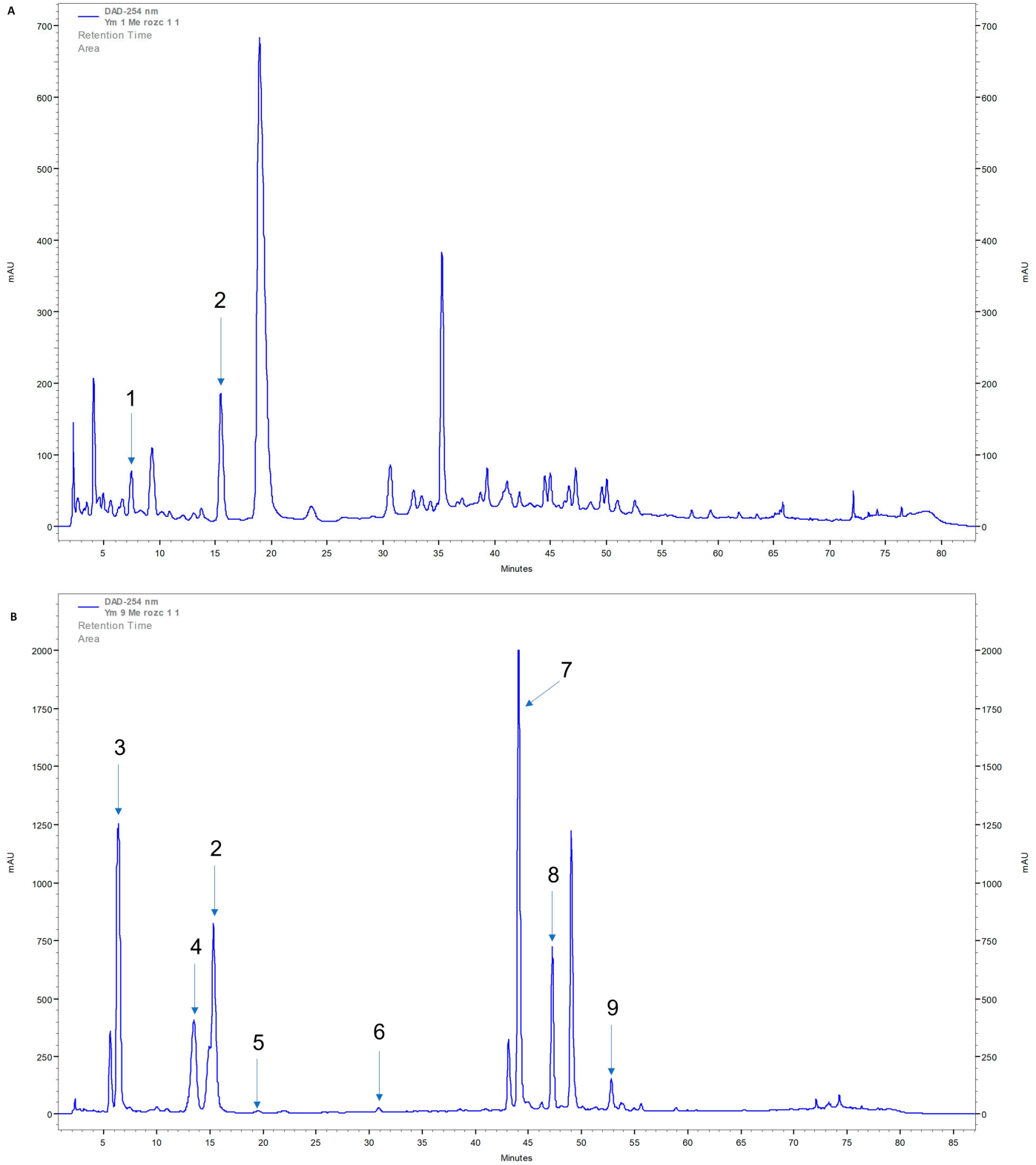

Figure 1. A representative chromatogram of the analyzed *Ilex paraguariensis* product samples: (**A**)—the methanol extract from sample P-1 (1—protocatechuic acid, 2—caffeine); (**B**)—the methanol extract from sample P-9 (2—caffeine, 3—neochlorogenic acid, 4—chlorogenic acid, 5—caffeic acid, 6—4-feruloylquinic acid, 7—isochlorogenic acid, 8—rutoside, 9—astragalin).

Table 3. Concentration of organic compounds in analyzed yerba mate products after methanol extraction. Each analysis was conducted three times. The letters next to values represent Tukey's HSD post hoc results, different letters (a–i) indicate significant differences between homogeneous groups of yerba mate products for each organic compound ($p < 0.05$).

Product	Protocatechuic Acid	Neochlorogenic Acid	Chlorogenic Acid	Caffeic Acid	4-Feruloylquinic Acid	Isochlorogenic Acid
			mg/g d.w.			
P-1	0.5493 ± 0.0153	-	-	-	-	-
P-2	-	2.6869 ± 0.1033 ^{f}	1.9787 ± 0.0210 ^{i}	1.7938 ± 0.1225 ^{a}	1.1955 ± 0.0095 ^{e}	1.7116 ± 0.0305 ^{g}
P-3	-	6.0020 ± 0.0505 ^{e}	4.6736 ± 0.0270 ^{h}	0.1709 ± 0.0051 ^{e}	0.830 ± 0.0047 ^{g}	5.9957 ± 0.0126 ^{f}
P-4	-	16.9913 ± 0.2093 ^{d}	14.7412 ± 0.0273 ^{a}	0.5362 ± 0.0036 ^{b}	2.4146 ± 0.0350 ^{a}	13.153 ± 0.2333 ^{e}
P-5	-	5.9032 ± 0.1702 ^{e}	5.6598 ± 0.0207 ^{g}	0.3493 ± 0.0073 ^{cd}	1.7904 ± 0.0374 ^{b}	2.6328 ± 0.1638 ^{g}
P-6	-	19.7390 ± 0.1671 ^{c}	11.3490 ± 0.0401 ^{b}	0.3132 ± 0.0152 ^{cd}	1.7453 ± 0.0233 ^{b}	20.6498 ± 0.0959 ^{c}
P-7	-	16.8391 ± 0.1578 ^{d}	9.1499 ± 0.0467 ^{f}	0.2566 ± 0.0031 ^{de}	1.0652 ± 0.0604 ^{f}	20.7914 ± 0.3593 ^{c}
P-8	-	17.6460 ± 0.1180 ^{d}	10.1720 ± 0.1032 ^{d}	0.1491 ± 0.0104 ^{e}	1.1760 ± 0.0330 ^{e}	16.2011 ± 0.1676 ^{d}
P-9	-	23.9750 ± 0.5405 ^{a}	9.6597 ± 0.0828 ^{e}	0.1881 ± 0.0139 ^{e}	1.3148 ± 0.0313 ^{d}	31.6961 ± 0.8483 ^{a}
P-10	-	21.1462 ± 0.6872 ^{b}	10.8487 ± 0.0828 ^{c}	0.3989 ± 0.0098 ^{c}	1.6283 ± 0.0321 ^{c}	27.5268 ± 0.6656 ^{b}

Table 4. Concentration of organic compounds in analyzed yerba mate products after water extraction. The letters next to values represent Tukey's HSD post hoc results, different letters (a–h) indicate significant differences between homogeneous groups of yerba mate products for each organic compound ($p < 0.05$).

Product	Protocatechuic Acid	Neochlorogenic Acid	Chlorogenic Acid	Caffeic Acid	4-Feruloylquinic Acid	Isochlorogenic Acid
			mg/g d.w.			
P-1	0.0433 ± 0.0012	-	-	-	-	-
P-2	-	0.4529 ± 0.0106 ^{h}	0.3570 ± 0.0144 ^{g}	0.4877 ± 0.0119 ^{a}	0.1993 ± 0.0035 ^{f}	0.4566 ± 0.0070 ^{f}
P-3	-	3.2747 ± 0.0401 ^{f}	2.4097 ± 0.1273 ^{e}	0.1129 ± 0.0013 ^{e}	0.6499 ± 0.0036 ^{c}	3.2666 ± 0.0209 ^{e}
P-4	-	10.2299 ± 0.1838 ^{a}	8.3120 ± 0.0335 ^{a}	0.4892 ± 0.0149 ^{a}	1.6579 ± 0.0197 ^{a}	8.2794 ± 0.0591 ^{a}
P-5	-	1.8603 ± 0.0312 ^{g}	1.6377 ± 0.0335 ^{f}	0.1172 ± 0.0033 ^{de}	0.5210 ± 0.0163 ^{d}	0.6586 ± 0.0188 ^{f}
P-6	-	6.5974 ± 0.2165 ^{e}	3.4445 ± 0.0919 ^{d}	0.1416 ± 0.0026 ^{c}	0.7419 ± 0.0199 ^{b}	4.5228 ± 0.1032 ^{d}
P-7	-	8.3190 ± 0.0983 ^{c}	3.9120 ± 0.1121 ^{c}	0.1357 ± 0.0029 ^{cd}	0.6834 ± 0.0039 ^{bc}	5.0671 ± 0.0623 ^{c}
P-8	-	8.9251 ± 0.0879 ^{b}	4.5349 ± 0.1444 ^{b}	0.1151 ± 0.0087 ^{de}	0.7066 ± 0.0390 ^{bc}	4.5225 ± 0.0612 ^{d}
P-9	-	7.0176 ± 0.2033 ^{d}	2.3529 ± 0.1161 ^{e}	0.0760 ± 0.0009 ^{f}	0.4587 ± 0.0314 ^{e}	4.9965 ± 0.1542 ^{c}
P-10	-	8.5851 ± 0.0664 ^{bc}	3.8017 ± 0.0558 ^{c}	0.2225 ± 0.0007 ^{b}	0.7121 ± 0.0251 ^{b}	7.6755 ± 0.1556 ^{b}

Flavonols that have been identified in *I. paraguariensis* are rutin (quercetin-3-*O*-rutinoside), quercetin (quercetin-3-*O*-glucoside), and nicotiflorin (kaempferol-3-*O*-rutinoside) [49]. Methanol and water extracts were analyzed for rutin content. Rutin ranged from 0.2420 to 5.5606 mg/g d.w. in water extracts and from 1.0018 to 8.5536 mg/g d.w. in methanol extracts.

Among the tested products, the dietary supplement in the form of P1 capsules was the only one found to contain protocatechuic acid, which was not present in any other tested products. Interestingly, this supplement lacked compounds characteristic of yerba mate, such as phenolic compounds, rutoside, and astragalin, which were present in other preparations. Additionally, the caffeine content was the lowest in this product.

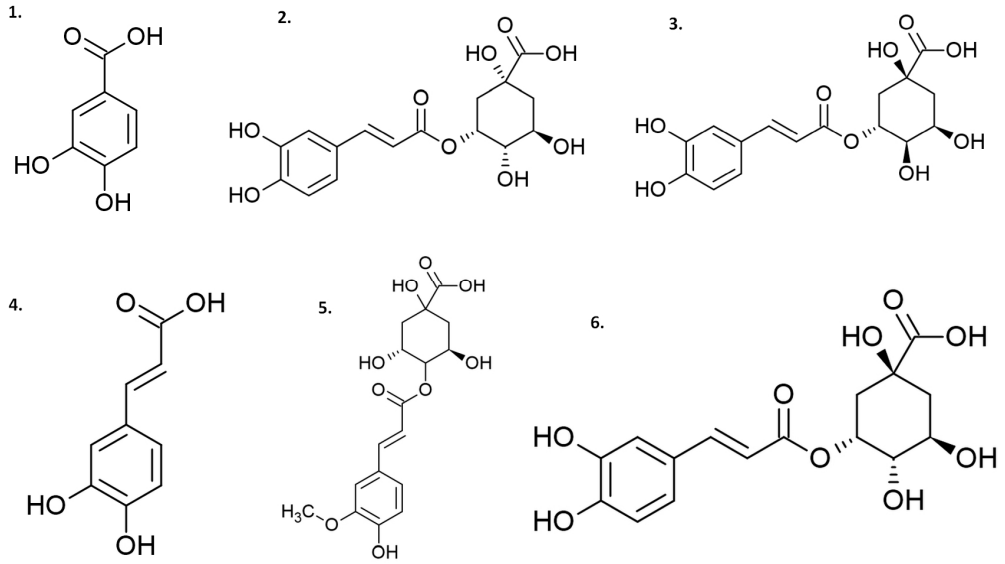

Figure 2. Structural formulas of bioactive substances analyzed in *Ilex paraguariensis* product samples: 1—protocatechuic acid, 2—neochlorogenic acid, 3—chlorogenic acid, 4—caffeic acid, 5—4-feruloylquinic acid, 6—isochlorogenic acid.

Table 5. Concentration of caffeine, rutoside and astragalin in P-1–P-10 products after methanol and water extraction. The letters next to values represent Tukey's HSD post hoc results, different letters (a–i) indicate significant differences between homogeneous groups of yerba mate products for each organic compound ($p < 0.05$).

Product	Caffeine (Methanol)	Rutoside (Methanol)	Astragalin (Methanol)	Caffeine (Water)	Rutoside (Water)	Astragalin (Water)
			mg/g d.w.			
P-1	0.1963 ± 0.0012 ^{g}	-	-	0.0278 ± 0.0005 ^{h}	-	-
P-2	3.4059 ± 0.0175 ^{a}	1.0018 ± 0.0033 ^{i}	0.2622 ± 0.0156 ^{h}	1.1400 ± 0.0073 ^{e}	0.2420 ± 0.0086 ^{f}	0.0768 ± 0.0006 ^{h}
P-3	0.5046 ± 0.0179 ^{f}	2.8098 ± 0.0706 ^{g}	0.8021 ± 0.0076 ^{f}	0.3884 ± 0.0072 ^{e}	1.9329 ± 0.0110 ^{d}	0.6239 ± 0.0030 ^{c}
P-4	0.8216 ± 0.0131 ^{d}	7.7982 ± 0.0614 ^{b}	0.9572 ± 0.0126 ^{e}	0.5952 ± 0.0067 ^{b}	5.5606 ± 0.0673 ^{e}	0.7593 ± 0.0038 ^{a}
P-5	0.5299 ± 0.007 ^{f}	1.4081 ± 0.1060 ^{h}	0.3596 ± 0.0050 ^{g}	0.1948 ± 0.0034 ^{g}	0.3773 ± 0.0392 ^{f}	0.1147 ± 0.0013 ^{g}
P-6	0.8693 ± 0.0027 ^{c}	6.6904 ± 0.1162 ^{d}	1.3451 ± 0.0619 ^{b}	0.4182 ± 0.0108 ^{d}	2.4442 ± 0.0519 ^{c}	0.5554 ± 0.0237 ^{d}
P-7	0.7347 ± 0.0043 ^{e}	4.7066 ± 0.0686 ^{f}	1.2064 ± 0.0055 ^{c}	0.4133 ± 0.0106 ^{d}	1.9574 ± 0.0173 ^{d}	0.6121 ± 0.0044 ^{c}
P-8	0.7254 ± 0.0162 ^{e}	5.6088 ± 0.0890 ^{e}	1.4552 ± 0.0234 ^{a}	0.5163 ± 0.0071 ^{c}	2.3780 ± 0.0838 ^{c}	0.7266 ± 0.0119 ^{b}
P-9	0.8638 ± 0.0056 ^{cd}	7.2466 ± 0.1062 ^{c}	1.1738 ± 0.0208 ^{c}	0.2986 ± 0.0033 ^{f}	1.2196 ± 0.0420 ^{e}	0.2801 ± 0.0033 ^{f}
P-10	1.0316 ± 0.0385 ^{b}	0.0465 ± 0.0465 ^{e}	1.0561 ± 0.0666 ^{d}	0.5194 ± 0.0054 ^{c}	2.8974 ± 0.0864 ^{b}	0.4340 ± 0.0045 ^{e}

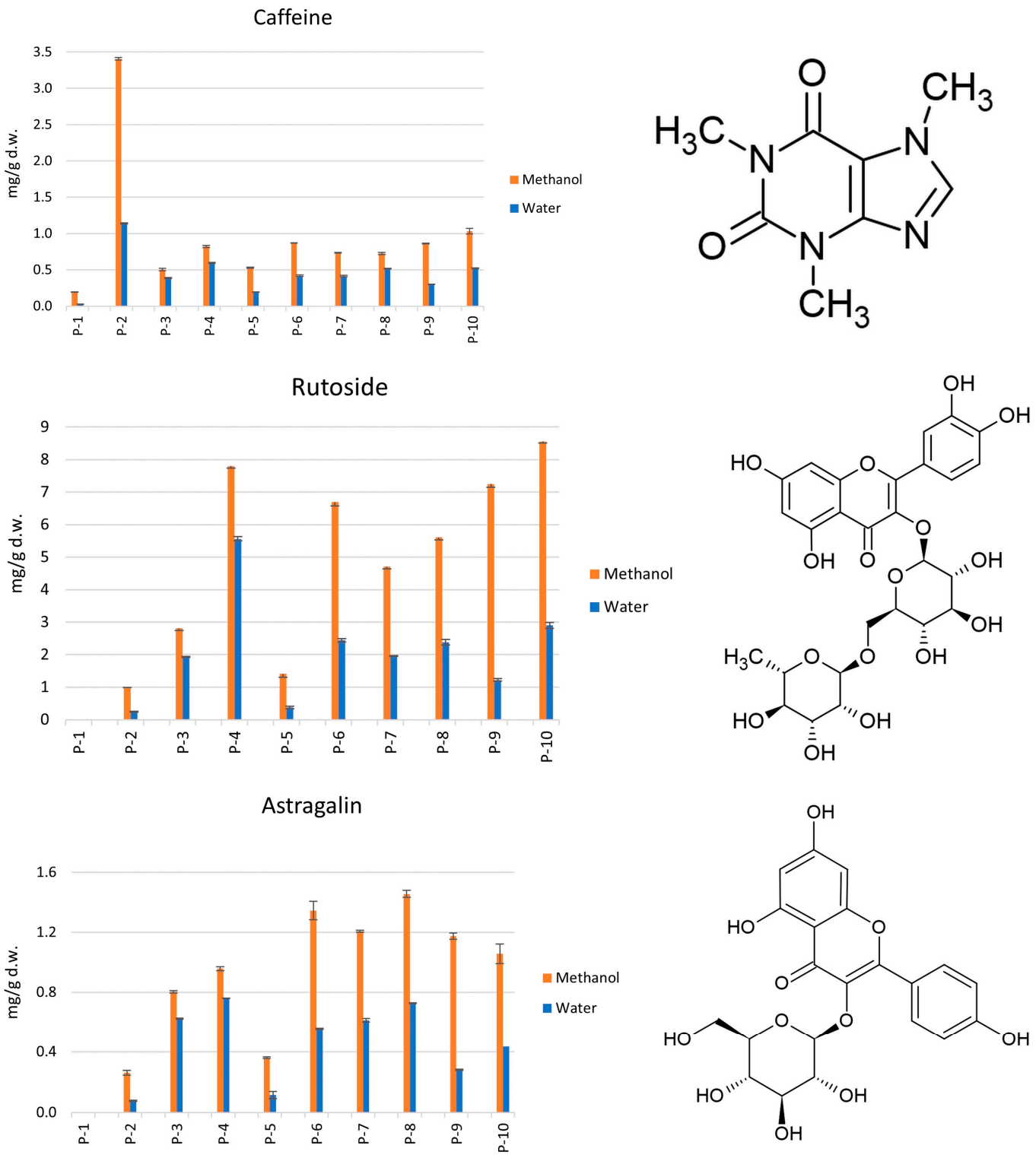

Figure 3. Concentration of caffeine, rutoside and astragalin in P-1–P-10 products [mg/g].

5. Conclusions

It seems reasonable to consider the potential of *I. paraguariensis* as a source of bioactive compounds for developing new products in the pharmaceutical, nutraceutical, and food sectors. It should also be noted that the options for dietary supplements containing solely yerba mate extract are limited in the market, and unfortunately, none of these offerings are of high quality. Among these supplements, the P-3 product, a yerba mate extract powder, stands out as the best. Interestingly, when comparing food supplements to raw

materials, the supplements had a lower content of investigated elements and compounds. Among the analyzed elements, magnesium was found to be the most abundant. The highest content of organic compounds with antioxidant properties (phenolic compounds, rutoside, astragalin, and caffeine) was determined in methanol extracts from instant yerba mate products and raw yerba mate materials. Therefore, the traditional preparation of yerba mate as a water infusion does not fully utilize the potential of the raw material. Furthermore, the determined polyphenol content did not match the figures declared by the producers, if stated on the packaging. The research suggests that drinking yerba mate infusions or opting for the soluble form of yerba mate could be a better choice than relying solely on food supplements. Specifically, consuming about 1 L of *I. paraguariensis* infusion per day would significantly contribute to meeting the recommended daily requirements for magnesium or polyphenols.

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